SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	BE-CSE (Final Year)	**Date:**	28/10/2026

Code of Conduct

I Maredi Neela Meghana 192311002 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **MAREDI NEELA MEGHANA**

QUESTIONS: 5 **Total Marks: 50**

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

ANSWER

Introduction

Grey-level transformation is a point-processing technique in which the intensity of each pixel in an image is changed according to a mathematical function. It is useful when an X-ray image is too dark because the bones, organs, and other anatomical structures may not be clearly visible. A suitable choice for this case is a linear contrast stretching transformation, which increases the range of intensity values and can also brighten the image.

1. Selected transformation

For an 8-bit grayscale image, pixel intensities range from 0 to 255. A linear transformation can be written as $s = a\,r + b$, where r is the original pixel intensity, s is the transformed intensity, a controls contrast, and b controls brightness. The output is clipped to the valid range 0–255. To brighten a dark X-ray, a positive value of b is used. To improve contrast as well, a can be chosen greater than 1, provided the output does not become excessively saturated.

2. Example calculation

Assume a dark X-ray has representative pixel values between 20 and 120. For illustration, use $s = 1.5r + 20$. This example transformation increases brightness by adding 20 and expands contrast by multiplying intensity by 1.5. The calculated values are rounded to the nearest integer and clipped to 255 if necessary.

3. Transformed intensity values

Original intensity (r)	Calculation	Transformed intensity (s)
20	1.5(20) + 20	50
40	1.5(40) + 20	80
60	1.5(60) + 20	110
80	1.5(80) + 20	140
100	1.5(100) + 20	170
120	1.5(120) + 20	200

4. Interpretation of the result

A pixel with original intensity 20 becomes 50, so very dark areas become easier to distinguish from the black background. An intensity of 60 becomes 110, making medium-gray anatomical details brighter. A pixel of intensity 100 becomes 170, increasing the difference between it and darker pixels. The intensity difference between original values 40 and 80 is 40; after transformation, their values are 80 and 140, a difference of 60. Thus, contrast between those structures is increased.

5. Practical application to X-ray images

The enhancement process should be applied to the grayscale image after checking the intensity histogram. If most pixels are concentrated near the lower end, a brightness adjustment and contrast stretching can make the useful intensity range occupy more of the available display range. The result should be reviewed by a trained medical professional. Image enhancement improves visibility, but it does not create anatomical information that was not captured, and it must not be treated as a diagnosis.

Advantages and limitations

The method is simple, fast, and easy to implement. It can improve the visibility of bones and soft-tissue boundaries in underexposed images. However, excessive brightness may make dark regions look washed out, while excessive contrast can cause bright regions to saturate at 255. Noise may also become more visible because contrast stretching affects noise pixels as well as useful details. The transformation parameters should therefore be selected carefully and checked against the original image.

Conclusion

Linear grey-level transformation is a suitable first step for enhancing a dark X-ray. The example $s = 1.5r + 20$ demonstrates how pixel values are brightened and spread apart. By improving the differences between neighboring intensity levels, the technique makes important anatomical structures easier to observe while requiring careful control to avoid loss of detail.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

ANSWER

Introduction

Night-time surveillance images often contain two problems: low contrast and random noise. Low contrast makes people and objects difficult to separate from the background, while noise creates unwanted bright and dark pixels. Histogram equalization and spatial filtering address these problems in different ways. Histogram equalization improves the distribution of image intensities, whereas a spatial filter processes pixels in a local neighborhood to reduce noise.

1. Histogram equalization

A histogram counts how many pixels occur at each grayscale intensity. In a low-contrast image, many pixels may be clustered within a narrow range, such as 40–100, even though the full 0–255 range is available. Histogram equalization uses the cumulative distribution of intensities to map the original values across a wider range. For L possible gray levels, the mapping is commonly expressed as $s_k = (L-1) \times CDF(r_k)$, where CDF is the cumulative distribution function. For an 8-bit image, L = 256.

2. Illustrative histogram mapping

Suppose a pixel intensity of 60 has a cumulative distribution value of 0.25, while intensity 100 has a cumulative distribution value of 0.70. The approximate equalized values are $255 \times 0.25 = 63.75$, or 64, and $255 \times 0.70 = 178.5$, or 179. These values are illustrative; actual mappings depend on the histogram of the captured surveillance image. The operation can make differences between people, clothing, vehicles, and the background more noticeable.

3. Spatial filtering to reduce noise

After contrast enhancement, a median filter is a useful choice when the image contains isolated bright or dark specks. For each pixel, the filter examines a small neighborhood, such as 3×3, sorts its intensity values, and replaces the center pixel with the median. Unlike a simple mean filter, the median is less affected by extreme values and tends to preserve edges better. For approximately Gaussian noise, a small Gaussian smoothing filter is another option, but it may blur fine edges.

4. Example of median filtering

Consider a 3 × 3 neighborhood with values 50, 51, 49, 52, 240, 50, 48, 53, and 51. The value 240 is a possible noise spike. When sorted, the middle (fifth) value is 51, so the center value 240 is replaced by 51. This removes the isolated spike while retaining the surrounding local intensity. The same operation is repeated across the image, with suitable handling at image borders.

5. Comparison of both techniques

Technique	Main operation	Benefit	Limitation
Histogram equalization	Remaps intensities using cumulative histogram	Improves overall contrast	May amplify noise or over-enhance some areas
Median filter	Replaces each pixel with neighborhood median	Removes isolated impulse noise and preserves edges reasonably well	Large windows can remove small details

6. Recommended processing sequence

First, inspect the original image and its histogram. Apply histogram equalization or, where global equalization produces too much contrast in some regions, contrast-limited adaptive histogram equalization (CLAHE) to enhance local contrast. Next, use a small median filter to remove isolated noise. Then inspect the image at normal viewing size to ensure that faces, outlines, and object boundaries remain visible. If filtering is performed after contrast enhancement, it can suppress noise that has also become more visible; however, severe noise may justify light denoising before equalization.

Benefits and limitations

Histogram equalization is computationally efficient and can reveal details in dark scenes, but global equalization may over-enhance noise or make some regions look unnatural. Median filtering removes impulse noise effectively, but a large neighborhood may erase thin features or small objects. Neither technique can recover details that were completely lost because of poor lighting, motion blur, or sensor limitations. Parameters should be selected for the camera and scene.

Conclusion

Histogram equalization improves the spread of intensity values, and median filtering reduces local noise. Used together and in a suitable order, they can make surveillance images clearer and improve the visibility of people and objects without relying on either technique alone.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

ANSWER

Introduction

Satellite images are used to study land cover, rivers, roads, forests, agricultural fields, and urban growth. Noise and blurred boundaries can make these regions difficult to distinguish. Frequency-domain filtering offers a way to separate slowly varying image information from rapid intensity changes. Low-pass filtering suppresses high-frequency components to smooth the image, while high-pass filtering emphasizes rapid changes that commonly occur at edges.

1. Frequency-domain representation

An image can be represented by its two-dimensional Fourier transform $F(u,v)$. Low spatial frequencies generally correspond to slowly changing brightness and broad regions, whereas high spatial frequencies often correspond to fine details, sharp boundaries, and noise. Filtering is performed by multiplying the Fourier transform by a filter function $H(u,v)$: $G(u,v) = H(u,v)F(u,v)$. The inverse Fourier transform of G produces the filtered image $g(x,y)$.

2. Low-pass filtering for smoothing

A low-pass filter allows low-frequency components to pass and attenuates high-frequency components. A Gaussian low-pass filter is often suitable because it reduces abrupt changes smoothly and produces fewer ringing artifacts than an ideal sharp-cutoff filter. In the frequency domain, a Gaussian low-pass filter can be represented as $H_{LP}(u,v) = \exp[-D^2(u,v)/(2D_0^2)]$, where $D(u,v)$ is the distance from the frequency origin and D_0 controls the cutoff scale.

3. Effect of the low-pass filter

Small random fluctuations and fine-grained noise are reduced, producing a smoother image. Broad regions such as large fields or water bodies may become more uniform. This can make overall land-cover patterns easier to interpret. However, smoothing also reduces high-frequency information, so narrow roads, small streams, and sharp boundaries may become blurred. The cutoff must therefore be selected to balance noise reduction and preservation of detail.

4. High-pass filtering for sharpening

A high-pass filter attenuates low-frequency components and retains or emphasizes higher-frequency components. It can highlight boundaries between land types, coastlines, river edges, roads, and transitions between vegetation and bare soil. A common sharpening approach is to combine the original image with a high-pass detail image: $g_{sharp}(x,y) = f(x,y) + k \cdot h_{HP}(x,y)$, where k controls the sharpening strength. The detail image contains rapid intensity changes.

5. Example of edge emphasis

Consider adjacent land regions with grayscale values 70 and 150. Their intensity difference is 80, which indicates a strong transition. A high-pass operation responds to this change and produces a stronger edge response around the boundary. In contrast, a uniform region with values close to 100 has little rapid variation and produces a weaker high-pass response. The exact numerical output depends on the selected filter and its coefficients.

6. Comparison of filters

Feature	Low-pass filter	High-pass filter
Main purpose	Smoothing and noise reduction	Edge enhancement and sharpening
Frequency response	Retains low frequencies; attenuates high frequencies	Attenuates low frequencies; retains/emphasizes high frequencies
Visual effect	Smoother regions; softer boundaries	Stronger boundaries and fine details
Main risk	Blurring small features	Amplifying noise and creating false edges

7. Practical workflow for satellite images

A suitable workflow is to inspect the original image, apply mild low-pass filtering if noise is significant, and then use a high-pass or unsharp-masking step to improve edge visibility. It is often better to sharpen a lightly smoothed image than to sharpen a noisy original, because high-pass filters can amplify noise. The filtered result should be compared with the original image, and important features should be checked at more than one scale.

Limitations and interpretation

Low-pass filtering may remove small but meaningful features, while high-pass filtering may emphasize unwanted noise and create artificial-looking edges if the sharpening strength is too high. Frequency-domain filtering also requires correct transform handling and careful interpretation of the frequency representation. Filtered images are aids for interpretation, not substitutes for validation with ground data or other satellite bands.

Conclusion

Low-pass filtering is primarily used to smooth images and reduce high-frequency noise. High-pass filtering is used to emphasize edges and sharpen boundaries. In satellite imagery, a controlled combination can improve the readability of geographical regions while preserving as much meaningful detail as possible.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

ANSWER

Introduction

In automated manufacturing, metal components must be inspected for cracks, scratches, pits, and other surface defects. Variations in illumination can make these defects difficult to separate from the background. Edge detection identifies rapid intensity changes that may correspond to crack boundaries, while thresholding converts a grayscale image into a binary image so that candidate defect pixels can be isolated and measured.

1. Image preparation

The captured image should first be converted to grayscale if it is in color. Uneven illumination may be corrected using background normalization or local contrast enhancement. A small amount of Gaussian smoothing can reduce sensor noise before edge detection. This preparation is important because noise and reflections can produce false edges that look similar to defects.

2. Edge detection using Canny

The Canny edge detector is appropriate when reliable boundaries are required. Its main steps are: (1) smooth the image with a Gaussian filter; (2) calculate intensity gradients; (3) determine edge strength and direction; (4) apply non-maximum suppression to make edges thin; and (5) use double thresholds and hysteresis to retain connected edges. Strong edges are accepted, while weaker edges are kept when connected to strong ones. This helps detect thin crack boundaries while limiting some isolated noise responses.

3. Thresholding for binary segmentation

Thresholding classifies each pixel according to its intensity. For a grayscale image f(x,y), a binary mask can be defined as $B(x,y) = 1$ when $f(x,y) < T$ and $B(x,y) = 0$ otherwise, if defects are darker than the surrounding metal. If defects are brighter, the inequality can be reversed. The threshold T should be selected from the image histogram or estimated automatically using a method such as Otsu's thresholding when the intensity distribution is suitable.

4. Example of binary classification

Assume an illustrative threshold T = 80 and that cracks appear darker than the metal surface. Pixels with intensities 35, 52, and 74 are classified as defect candidates (1), while pixels with intensities 95, 130, and 210 are classified as background (0). This produces a binary image where candidate defect regions are white and the background is black, depending on the chosen display convention. These values are examples; a real threshold must be calibrated using the camera, material, and lighting conditions.

5. Combining edges and thresholding

Edge detection and thresholding provide complementary information. Canny detects the likely outlines of cracks, while thresholding identifies pixels belonging to dark or bright defect regions. The edge map can be used to check or refine the boundaries of the thresholded regions. Morphological operations, such as removing very small isolated components or closing short gaps, can improve the mask. These operations should be used carefully so that narrow cracks are not accidentally removed.

6. Inspection workflow

Stage	Method	Output / purpose
1	Grayscale conversion and mild smoothing	Reduce color complexity and noise
2	Canny edge detection	Thin candidate crack boundaries
3	Thresholding / Otsu threshold	Binary mask of candidate defect pixels
4	Morphological cleanup and component analysis	Join small gaps and measure candidate defects
5	Validation against quality limits	Flag components for acceptance or review

7. Evaluation and decision

The system can measure candidate defect area, length, width, and position from the binary mask. For example, the number of foreground pixels gives an estimate of the projected defect area when pixel calibration is known. A component can be flagged for review if its measured defect area or crack length exceeds a validated limit. The threshold should be developed using labelled examples of acceptable and defective components and tested on separate images.

Advantages and limitations

Canny provides thin, connected edge responses and is less sensitive to noise than a simple gradient operator when its smoothing and thresholds are selected well. Thresholding is easy to interpret and produces a binary mask suitable for measurements. However, uneven illumination, shiny reflections, scratches that are not defects, and very low-contrast cracks may cause false positives or missed defects.

A fixed threshold may fail when brightness changes substantially across the image, so adaptive thresholding or illumination correction may be needed.

Conclusion

Canny edge detection locates candidate crack boundaries, and thresholding separates likely defective pixels from the background. Together, they create a practical basis for binary defect masks and automatic measurements. Careful calibration and validation are necessary before the system is used for production decisions.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

ANSWER

Introduction

Agricultural aerial images contain different land regions, such as crop fields, water bodies, bare soil, and built-up areas. Some regions may have similar brightness, and shadows or image noise may break their boundaries. Region-based segmentation groups connected pixels that share similar properties, while edge linking and boundary detection help identify the outlines of the resulting regions.

1. Region-based segmentation

Region-based segmentation divides an image into connected regions according to a similarity criterion. Similarity may be based on grayscale intensity, color, texture, or vegetation indices. A common approach is region growing. The method starts with one or more seed pixels and adds neighbouring pixels when their values are sufficiently similar to the current region. Growth stops when neighbouring pixels fail the similarity test or when a region boundary is reached.

2. Region-growing procedure

First, select seed points inside known examples of a crop field, water body, or other target region. Next, examine neighbouring pixels using 4-connectivity or 8-connectivity. For a grayscale example, a pixel can be added when its intensity differs from the region mean by less than a chosen tolerance T. The region mean is updated as pixels are added. Finally, label connected regions and remove only those tiny regions that are known to be irrelevant, taking care not to erase small genuine fields or water channels.

3. Illustrative similarity rule

Suppose a seed region has an average grayscale intensity of 90 and a tolerance T = 12. A neighbouring pixel with intensity 98 differs by 8 and can be included. A pixel with intensity 108 differs by 18 and would not be included under this simple rule. The values are illustrative. In real aerial images, color or texture features may separate crop and water regions more reliably than grayscale intensity alone.

4. Edge linking

An edge detector may find only portions of a field boundary because of shadows, low contrast, vegetation, or sensor noise. Edge linking connects edge fragments that are close to one another and have compatible directions or gradient strengths. Nearby endpoints can be linked when the gap is small and their orientations are consistent. This creates longer, more continuous boundary curves and reduces fragmentation.

5. Boundary detection

Once regions are labelled, a boundary can be found by checking pixels whose neighbouring pixels have a different region label. The boundary forms a contour around each connected region. Region labels and edge maps can be combined: the region segmentation identifies which pixels belong together, while the linked edges provide evidence for the location of the border. If the two disagree, the system can apply a refinement step or mark the region for manual review.

6. Example of a combined workflow

Stage	Technique	Purpose
1	Choose seed pixels and similarity rule	Start regions in known land types
2	Region growing	Group connected, similar neighbouring pixels
3	Edge detection and edge linking	Connect discontinuous boundary fragments
4	Boundary detection from labels and edges	Trace the outline of each region
5	Area calculation and validation	Support field and water-body mapping

7. Benefits for agricultural monitoring

The segmented image can estimate the area of crop fields, identify water bodies, and support comparisons across dates. If the ground sampling distance is known, the number of pixels in a region can be converted into an approximate physical area. More consistent boundaries can improve field-area estimates and help monitor changes caused by flooding, crop growth, or land-use changes. Using the same processing settings across dates can support fair comparisons, although seasonal and lighting changes still need to be considered.

Limitations and improvements

Region growing may leak across boundaries when neighbouring regions have similar intensities. Results also depend on seed placement and the similarity threshold. Edge linking can accidentally connect unrelated edge fragments if the distance or orientation criteria are too relaxed. These problems can be reduced by using color and texture features, multiple seed points, adaptive thresholds, vegetation indices, and validation against field observations or reliable map data.

Conclusion

Region-based segmentation groups neighbouring pixels with similar characteristics, while edge linking repairs broken edge fragments and boundary detection traces the outlines of regions. Their combination can produce more complete maps of crop fields, water bodies, and surrounding land, improving the usefulness of aerial imagery for agricultural monitoring.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
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